WEEK-15 ASSESSMENT

Step 1: Database First (EF Core)

- Created the NorthwindCatalog project using Entity Framework Core Db First approach and scaffolded the Northwind database to generate entity classes automatically using following command.

NothwindCatalog.Services | NorthwindCatalog.Web

```
<Project Sdk="Microsoft.NET.Sdk.Web">

    <PropertyGroup>
        <TargetFramework>net8.0</TargetFramework>
        <Nullable>enable</Nullable>
        <ImplicitUsings>enable</ImplicitUsings>
    </PropertyGroup>
    <ItemGroup>
        <PackageReference Include="AutoMapper" Version="16.1.1" />
        <PackageReference Include="Microsoft.EntityFrameworkCore.Design" Version="8.0.23">
            <PrivateAssets>all</PrivateAssets>
            <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive</IncludeAssets>
        </PackageReference>
        <PackageReference Include="Microsoft.VisualStudio.Web.CodeGeneration.Design" Version="8.0.23" />
        <PackageReference Include="Microsoft.EntityFrameworkCore.SqlServer" Version="8.0.1" />
        <PackageReference Include="Microsoft.EntityFrameworkCore.Tools" Version="8.0.23">
            <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive</IncludeAssets>
            <PrivateAssets>all</PrivateAssets>
        </PackageReference>
        <PackageReference Include="Swashbuckle.AspNetCore" Version="6.6.2" />
    </ItemGroup>
</Project>
```

106 % | No issues found | Ln: 21, Ch: 14, Col: 17 | TABS | CRLF | UTF-8

appsettings.json* | NorthwindCatalog.Services* | NorthwindCat...ces: Overview

Schema: <No Schema Selected>

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
  "ConnectionStrings": {
    "DefaultConnection": "Server=.\\sqlexpress;Database=Northwind;Trusted_Connection=True;TrustServerCertificate=True;"
  }
}
```

79 % | No issues found | Ln: 13, Ch: 1 | SPC | CRLF | UTF-8

Solution Explorer

Search Solution Explorer (Ct

Solution 'NorthwindCatalog' (

NorthwindCatalog.Service

- Connected Services
- Dependencies
- Properties
- Controllers
- Data
  - NorthwindContext.cs
- Models
  - Category.cs
  - Product.cs
- appsettings.json
- NorthwindCatalog.Servic
- Program.cs

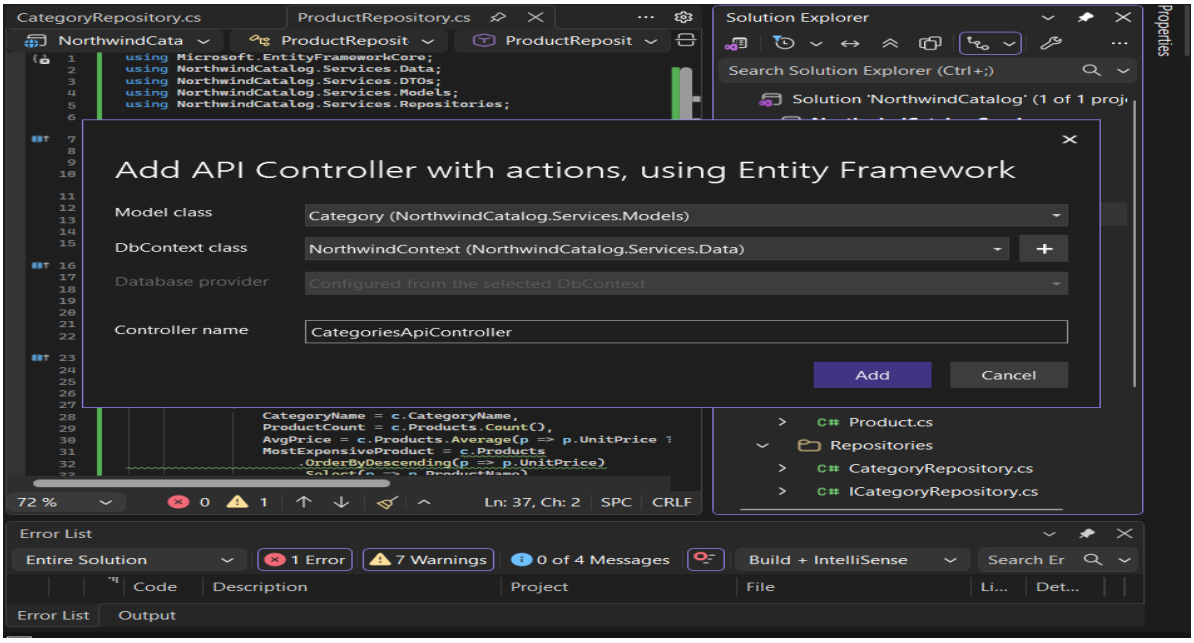
Developer PowerShell

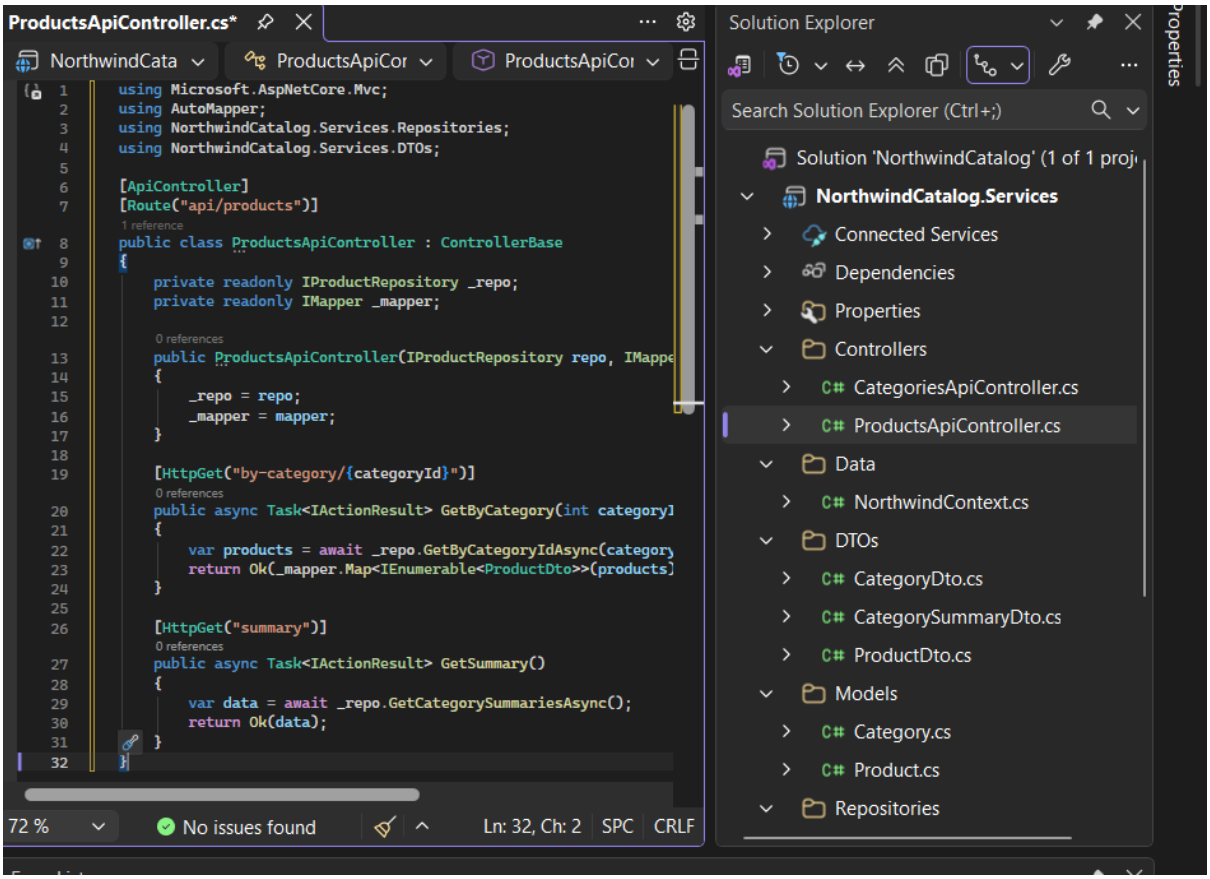
```
Build started...
Build succeeded.
To protect potentially sensitive information in your connection string, you should move it out of source code. You can avoid scaffolding the connection string by using the Name= syntax to read it from configuration - see https://go.microsoft.com/fwlink/?linkid=2131148. For more guidance on storing connection strings, see https://go.microsoft.com/fwlink/?LinkId=723263.
Skipping foreign key 'FK_Products_Suppliers' on table 'dbo.Products' since principal table 'dbo.Suppliers' was not found in the model. This usually happens when the principal table was not included in the selection set.
PS C:\Users\image\source\repos\NorthwindCatalog\NorthwindCatalog.Services>
```

- Fluent API Configuration and made API controllers also created repositories and DTOs

```
using Microsoft.EntityFrameworkCore;
using NorthwindCatalog.Services.Data;
using NorthwindCatalog.Services.Models;
using NorthwindCatalog.Services.Repositories;

2 references
public class CategoryRepository : ICategoryRepository
{
    private readonly NorthwindContext _context;

    0 references
    public CategoryRepository(NorthwindContext context)
    {
        _context = context;
    }

    2 references
    public async Task<IEnumerable<Category>> GetAllAsync()
    {
        return await _context.Categories.ToListAsync();
    }
}
```

```
NorthwindCatalog.Services

    protected override void OnConfiguring(DbContextOption
#warning To protect potentially sensitive information in
        => optionsBuilder.UseSqlServer("Server=.\\sqlexpr

    0 references
    protected override void OnModelCreating(ModelBuilder
    {
        modelBuilder.Entity<Category>(entity =>
        {
            entity.HasIndex(e => e.CategoryName, "Categor

            entity.Property(e => e.CategoryId).HasColumnN
            entity.Property(e => e.CategoryName).HasMaxLe
            entity.Property(e => e.Description).HasColumn
            entity.Property(e => e.Picture).HasColumnType
        });

        modelBuilder.Entity<Product>(entity =>
        {
            entity.HasIndex(e => e.CategoryId, "Categorie

            entity.HasIndex(e => e.CategoryId, "Category

            entity.HasIndex(e => e.ProductName, "ProductN

            entity.HasIndex(e => e.SupplierId, "SupplierI
```


CategoryRepository.cs
ProductRepository.cs
NorthwindCata
ProductReposit
using Microsoft.EntityFrameworkCore;
using NorthwindCatalog.Services.Data;
using NorthwindCatalog.Services.DTOs;
using NorthwindCatalog.Services.Models;
using NorthwindCatalog.Services.Repositories;
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'NorthwindCatalog' (1 of 1 proj
Add API Controller with actions, using Entity Framework
Model class
Category (NorthwindCatalog.Services.Models)
DbContext class
NorthwindContext (NorthwindCatalog.Services.Data)
Database provider
Configured from the selected DbContext
Controller name
CategoriesApiController
Add
Cancel
CategoryName = c.CategoryName,
ProductCount = c.Products.Count(),
AvgPrice = c.Products.Average(p => p.UnitPrice
MostExpensiveProduct = c.Products
.OrderByDescending(p => p.UnitPrice)
Product.cs
Repositories
CategoryRepository.cs
ICategoryRepository.cs
72 %
Ln: 37, Ch: 2
SPC
CRLF
Error List
Entire Solution
1 Error
7 Warnings
0 of 4 Messages
Build + IntelliSense
Code
Description
Project
File
Error List
Output


ProductsApiController.cs*
NorthwindCata
ProductsApiCor
using Microsoft.AspNetCore.Mvc;
using AutoMapper;
using NorthwindCatalog.Services.Repositories;
using NorthwindCatalog.Services.DTOs;
[ApiController]
[Route("api/products")]
1 reference
public class ProductsApiController : ControllerBase
{
private readonly IProductRepository _repo;
private readonly IMapper _mapper;
0 references
public ProductsApiController(IProductRepository repo, IMappe
{
_repo = repo;
_mapper = mapper;
}
[HttpGet("by-category/{categoryId}")]
0 references
public async Task<IActionResult> GetByCategory(int category
{
var products = await _repo.GetByCategoryIdAsync(category
return Ok(_mapper.Map<IEnumerable<ProductDto>>(products
}
[HttpGet("summary")]
0 references
public async Task<IActionResult> GetSummary()
{
var data = await _repo.GetCategorySummariesAsync();
return Ok(data);
}
}
72 %
No issues found
Ln: 32, Ch: 2
SPC
CRLF
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'NorthwindCatalog' (1 of 1 proj
NorthwindCatalog.Services
Connected Services
Dependencies
Properties
Controllers
CategoriesApiController.cs
ProductsApiController.cs
Data
NorthwindContext.cs
DTOs
CategoryDto.cs
CategorySummaryDto.cs
ProductDto.cs
Models
Category.cs
Product.cs
Repositories

THE APIS WORKING CORRECTLY:

Swagger UI

Ask Gemini

localhost:7184/swagger/index.html

CategoriesApi

GET /api/categories

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:7184/api/categories' \
  -H 'accept: */*'
```

Request URL

https://localhost:7184/api/categories

Server response

Code

Details

200

Response body


```
[
  {
    "categoryId": 1,
    "categoryName": "Beverages",
    "imageUrl": "/images/Beverages.jpg"
  },
  {
    "categoryId": 2,
    "categoryName": "Condiments",
    "imageUrl": "/images/Condiments.jpg"
  },
  {
    "categoryId": 3,
    "categoryName": "Confections",
    "imageUrl": "/images/Confections.jpg"
  },
  {
    "categoryId": 4,
    "categoryName": "Dairy Products",
    "imageUrl": "/images/Dairy Products.jpg"
  },
  {
    "categoryId": 5,
    "categoryName": "Grains/Cereals",
```


GET /api/products/by-category/{categoryId}

Parameters

Cancel

Name

Description

categoryId * required

integer($int32)

(path)

1

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:7184/api/products/by-category/1' \
  -H 'accept: */*'
```

Request URL

https://localhost:7184/api/products/by-category/1

Server response

Code

Details

200

Response body


```
[
  {
    "productName": "Chai",
    "unitPrice": 18,
    "unitsInStock": 39,
    "inventoryValue": 702
  },
  {
    "productName": "Updated Chang",
    "unitPrice": 300,
    "unitsInStock": 17,
    "inventoryValue": 5100
  },
  {
    "productName": "Guaraná Fantástica",
    "unitPrice": 4.5,
    "unitsInStock": 20,
    "inventoryValue": 90
  },
  {
    "productName": "Sasquatch Ale",
```

Parameters

Cancel

No parameters

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:7184/api/products/summary' \
  -H 'accept: */*'
```

Request URL

```
https://localhost:7184/api/products/summary
```

Server response

Code | Details

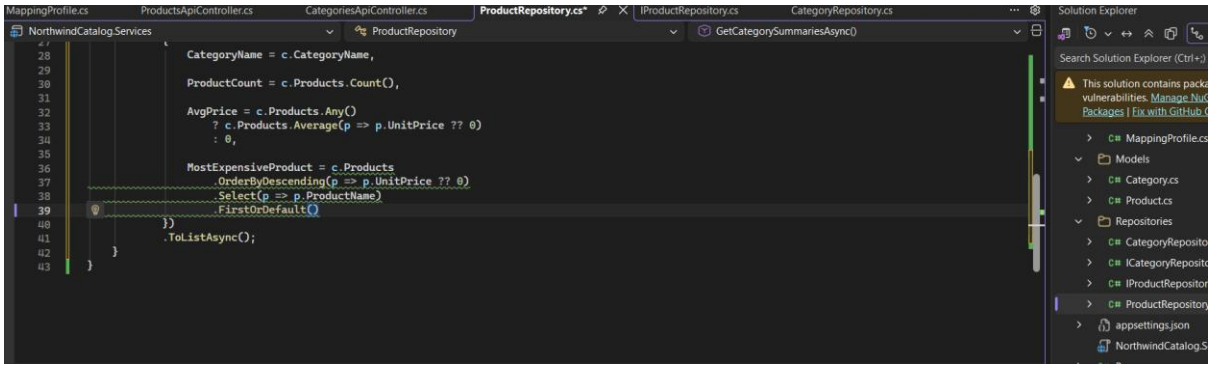
200

Response body


```
[
  {
    "categoryName": "Beverages",
    "productCount": 12,
    "avgPrice": 61.395833,
    "mostExpensiveProduct": "Updated Chang"
  },
  {
    "categoryName": "Condiments",
    "productCount": 12,
    "avgPrice": 23.0625,
    "mostExpensiveProduct": "Vegie-spread"
  },
  {
    "categoryName": "Confections",
    "productCount": 13,
    "avgPrice": 25.16,
    "mostExpensiveProduct": "Sir Rodney's Marmalade"
  },
  {
    "categoryName": "Dairy Products",
    "productCount": 10,
    "avgPrice": 28.73,
    "mostExpensiveProduct": "Raclette Courdavault"
  },
  {
    "categoryName": "Grains/Cereals",
    "productCount": 7,
```


Download

Step 2: MVC Layer where Http client was used (UI implementation)



File Edit View Git Project Build Debug Test Tools Extensions Window Northw...atalog SP

Help

Debug Any CPU

This project may be compatible for remote development. Would you like to install Visual Studio's Linux and Mac support? Install Linux and Mac Support More Info Do not ask again

SummaryController.cs CategoriesController.cs Program.cs*

NorthwindCatalog.Web SummaryController SummaryController(IHttpCli

```
using Microsoft.AspNetCore.Mvc;
using NorthwindCatalog.Services.DTOs;

1 reference
public class SummaryController : Controller
{
    private readonly HttpClient _client;

    0 references
    public SummaryController(IHttpClientFactory factory)
    {
        _client = factory.CreateClient("api");
    }

    0 references
    public async Task<IActionResult> Index()
    {
        var summary = await _client.GetFromJsonAsync<List<CategorySummar
            ("api/products/summary");

        return View(summary);
    }
}
```

96 % No issues found Ln: 8, Ch: 50 SPC CRLF UTF-8 with BOM

Solution Explorer

Search Solution Explore

- Solution 'NorthwindCata
- NorthwindCatalog.S
- NorthwindCatalog.We
  - Connected Services
  - Dependencies
  - Properties
  - wwwroot
  - Controllers
    - CategoriesContrc
    - HomeController.
    - ProductsControll
    - SummaryControl
  - Models
  - Views
  - appsettings.json
  - Program.cs

Properties

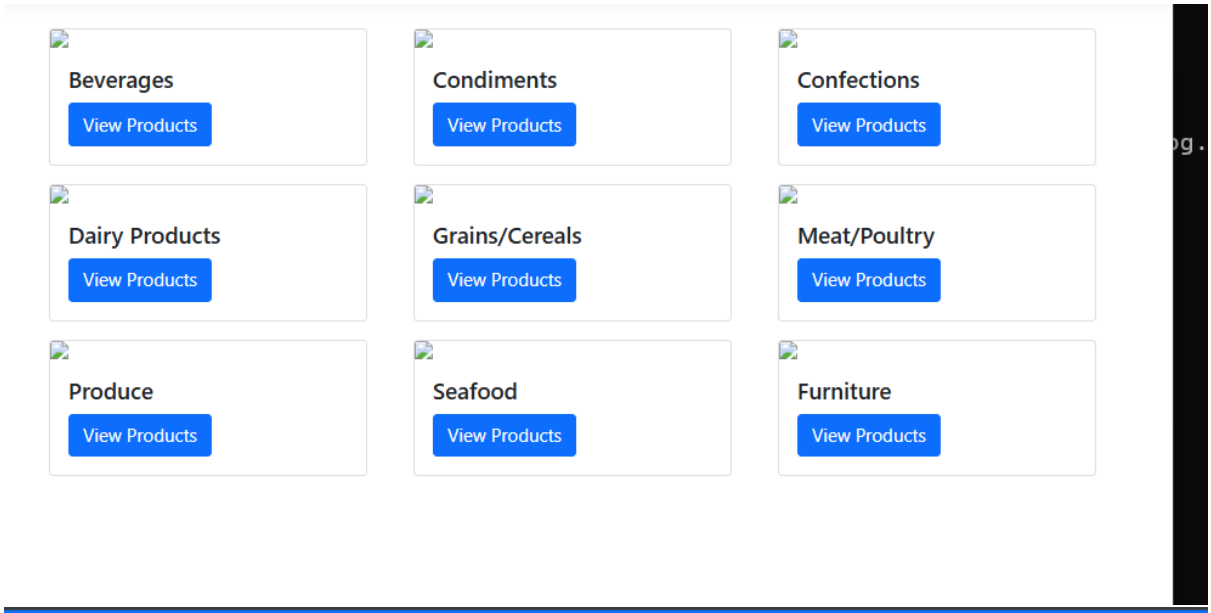
Output

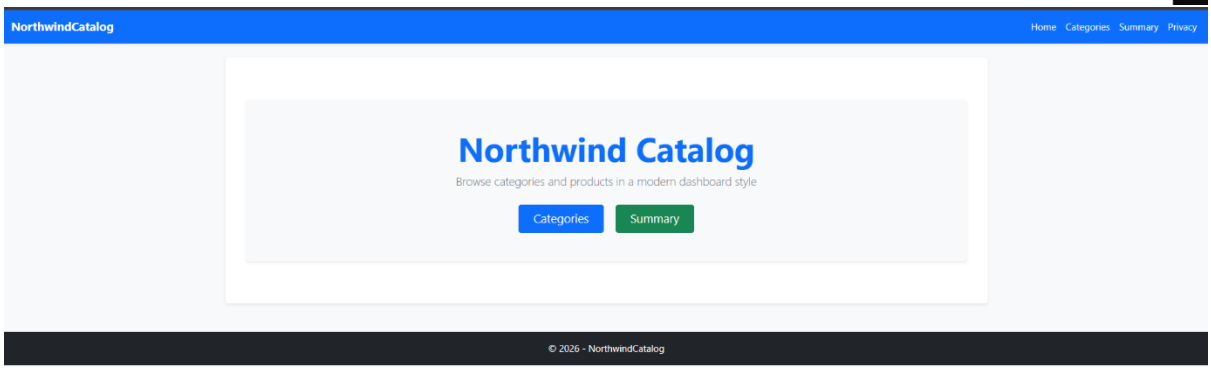
Show output from: Debug

```
    ORDER BY COALESCE([p2].[UnitPrice], 0.0) DESC) AS [MostExpensiveProduct]
FROM [Categories] AS [c]
The program '[9112] NorthwindCatalog.Services.exe' has exited with code 4294967295 (0xffffffff).
```

```
using Microsoft.AspNetCore.Mvc;
using NorthwindCatalog.Services.DTOs;

1 reference
public class SummaryController : Controller
{
    private readonly HttpClient _client;

    0 references
    public SummaryController(IHttpClientFactory factory)
    {
        _client = factory.CreateClient("api");
    }

    0 references
    public async Task<IActionResult> Index()
    {
        var summary = await _client.GetFromJsonAsync<List<CategorySummaryDto>>
            ("api/products/summary");

        return View(summary);
    }
}
```

```
@model IEnumerable<NorthwindCatalog.Services.DTOs.CategoryDto>

@if (Model == null || !Model.Any())
{
    <div class="alert alert-info">No categories available.</div>
}
else
{
    <div class="row">
        @foreach (var item in Model)
        {
            <div class="col-md-4">
                <div class="card m-2">
                    <img src="@item.ImageUrl" class="card-img-top" alt="@item.CategoryName" />
                    <div class="card-body">
                        <h5 class="card-title">@item.CategoryName</h5>
                        <a asp-controller="Products" asp-action="ByCategory" asp-route-id="@item.CategoryId" class="btn btn-primary">View Products</a>
                    </div>
                </div>
            </div>
        }
    </div>
}
```





Beverages
View Products

Condiments
View Products

Confections
View Products

Dairy Products
View Products

Grains/Cereals
View Products

Meat/Poultry
View Products

Produce
View Products

Seafood
View Products

NorthwindCatalog

Pavlova
Price: ₹ 17.45
Stock: 29
Inventory Value: ₹ 506.05

Teatime Chocolate Biscuits
Price: ₹ 9.20
Stock: 25
Inventory Value: ₹ 230.00

Sir Rodney's Marmalade
Price: ₹ 81.00
Stock: 40
Inventory Value: ₹ 3,240.00

Sir Rodney's Scones
Price: ₹ 10.00
Stock: 3
Inventory Value: ₹ 30.00

NuNuCa Nuß-Nougat-Creme
Price: ₹ 14.00
Stock: 76
Inventory Value: ₹ 1,064.00

Gumbär Gummibärchen
Price: ₹ 31.23
Stock: 15
Inventory Value: ₹ 468.45

Schoggi Schokolade
Price: ₹ 43.90
Stock: 49
Inventory Value: ₹ 2,151.10

Zaanse koeken
Price: ₹ 9.50
Stock: 36
Inventory Value: ₹ 342.00

Chocolade
Price: ₹ 12.75
Stock: 15
Inventory Value: ₹ 191.25

Maxilaku
Price: ₹ 20.00
Stock: 10
Inventory Value: ₹ 200.00

Valkoinen suklaa
Price: ₹ 16.25
Stock: 65
Inventory Value: ₹ 1,056.25

Tarte au sucre
Price: ₹ 49.30
Stock: 17
Inventory Value: ₹ 838.10

Scottish Longbreads
Price: ₹ 12.50

NorthwindCatalog

Category	Product Count	Avg Price	Most Expensive
Beverages	12	₹ 61.40	Updated Chang
Condiments	12	₹ 23.06	Vegie-spread
Confections	13	₹ 25.16	Sir Rodney's Marmalade
Dairy Products	10	₹ 28.73	Raclette Courdavault
Grains/Cereals	7	₹ 20.25	Gnocchi di nonna Alice
Meat/Poultry	6	₹ 54.01	Thüringer Rostbratwurst
Produce	5	₹ 32.37	Manjimup Dried Apples
Seafood	12	₹ 20.68	Carnarvon Tigers

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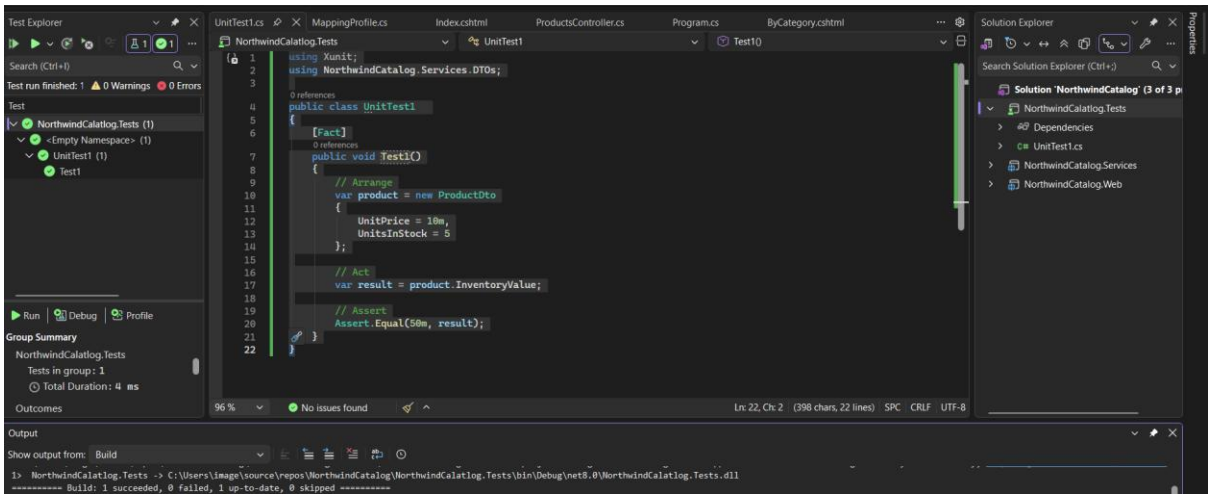
Step 3: Unit Testing (XUnit) - Created separate test project for unit testing